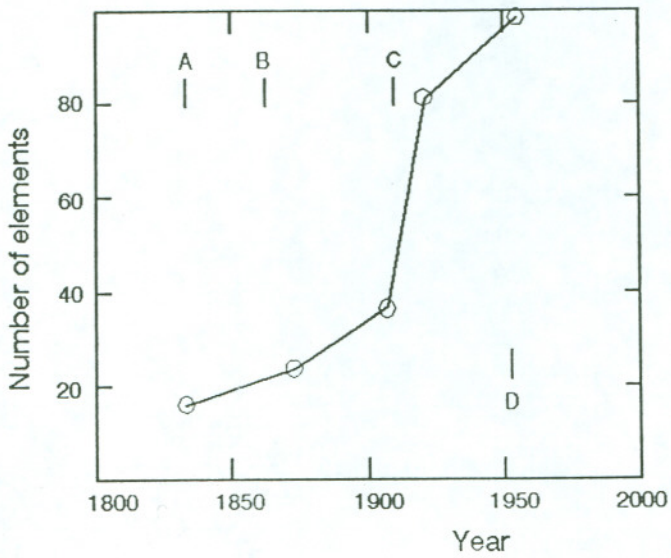
ANALYTICAL CUMULATIVE EXAMINATION

2 DECEMBER 1995



The figure above shows the number of elements known as a function of year, excluding the artificially made elements. The vertical marks indicate the invention of analytical instruments that played a major role in the history of the discovery of the elements. All of the techniques are in common use today.

(a) Identify the techniques indicated (5% per technique).

(b) Describe, as quantitatively as possible, one of the techniques (40%).

(c) Describe, as quantitatively as possible, any widely-used modern technique for quantitative elemental analysis not represented on the figure. State its advantages and disadvantages (40%).

In answering parts (b) and (c) credit will be given for:

completeness and technical accuracy
overall structure of essay
(i.e., use of headings, subheadings, introduction, middle and end)
appropriate use of figures and tables
appropriate use specific examples
appropriate use of equations and other formal treatments
good use of English